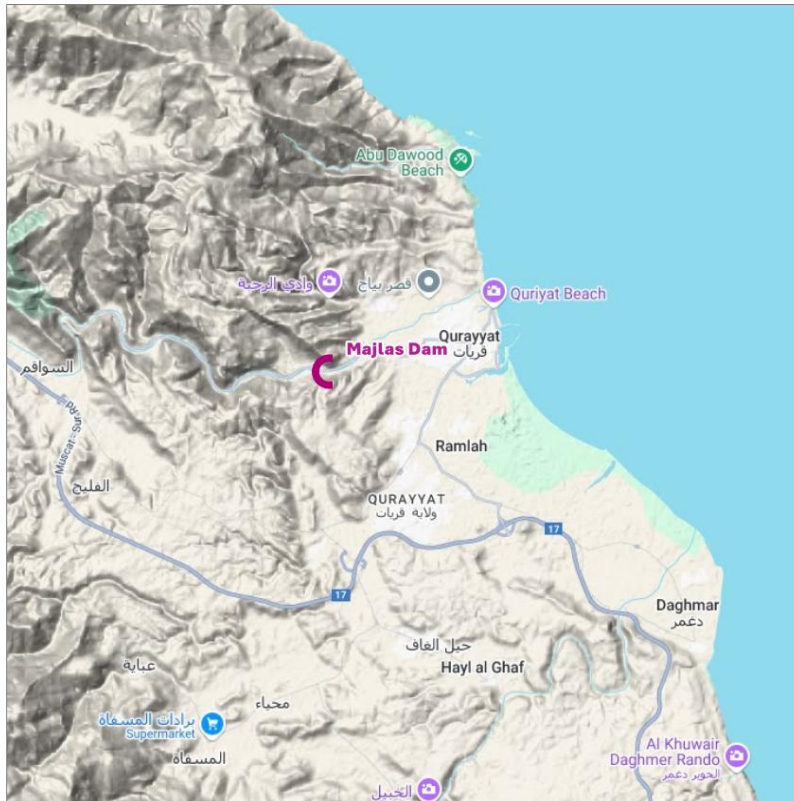




Sultanate of Oman
Ministry of Agriculture, Fisheries & Water Resources

Consultancy Services for Design Review and Supervision of Wadi Majlas Dam in Wilayat Qurayat, Muscat Governorate



Wadi Majlas Flood Protection Dam Hydrology Report

Rev 00
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EXECUTIVE SUMMARY

This report presents the updated hydrological study for the **Wadi Majlas Flood Protection Dam**, Sultanate of Oman. It supersedes the hydrology chapter of the previous design report and documents the rainfall data analysis, gap filling, regional frequency analysis, Probable Maximum Precipitation (PMP), rainfall-runoff modelling, calibration review and design flood estimates for the dam and the downstream dyked-channel protection scheme.

The catchment draining to the dam is about **571 km²** (604 km² including the downstream sub-catchments added for the dyked-channel design), rising from ~80 m at the dam to nearly 1,970 m in the Hajar mountains. A HEC-HMS model was built on a 5 m NSA DTM with an SCS curve-number loss method (GCN250 grid, catchment average CN about 89), Kirpich lag times and depth-area reduced 24-hour design storms.

Design rainfall was derived by a **regional frequency analysis following the Hosking & Wallis index-flood procedure, using the GEV distribution fitted by L-moments**. Annual maxima were extracted from the observed records and internal gaps filled by the normal-ratio method (single pass from observed data only; real zeros preserved). The region is statistically homogeneous (heterogeneity $H = -1.1$; no discordant sites). The local depths agree with the Oman Flood-Mapping design rainfall at short return periods (2-year 23.9 vs 27.6 mm) but produce a much heavier tail at rare return periods because the short records are dominated by the exceptional 2007 Cyclone Gonu; the Flood-Mapping depths, used to drive the HEC-HMS runs, were retained as the design basis.

A review of the previous calibration found that the observed Gonu hydrograph at the Majlass gauge could only be reproduced by reducing gauged rainfall to physically implausible levels, whereas the Hayfadh gauge was well reproduced. That observed hydrograph is judged erroneous and was excluded; the previous report did not flag it.

Design peak discharges were produced for return periods from 2 to 10,000 years plus the PMP, with and without the Majlas reservoir. The reservoir stores the full inflow up to about the 200-year event and attenuates the 10,000-year peak from about 14,170 to 8,750 m³/s (~38%).



1. INTRODUCTION

The Ministry of Agriculture, Fisheries and Water Resources is developing a flood-protection scheme for Wadi Majlas, in Wilayat Qurayat (Muscat Governorate), comprising a flood-detention dam and a downstream dyked (leveed) channel. This report updates the hydrological basis of the scheme. Its objectives are to:

- characterise the Wadi Majlas catchment and its hydro-meteorological setting;
- analyse the available rainfall records (quality control, gap filling) and derive design rainfall for durations to 24 hours and return periods to 10,000 years, and estimate the PMP;
- build and calibrate a HEC-HMS rainfall-runoff model of the catchment and downstream reaches;
- review the calibration adopted in the previous study against the observed cyclone hydrographs; and
- produce design flood hydrographs and peak discharges for the dam and the downstream protection works.

Two HEC-HMS models were developed: an initial model used to construct and calibrate the catchment response against observed hydrographs and peak flows, and a final model that adds the downstream sub-catchments for the dyked-channel design and applies a depth-area analysis to represent the areal reduction of extreme rainfall.



2. STUDY AREA AND PHYSIOGRAPHY

The Wadi Majlas catchment drains the eastern Hajar mountains towards the dam site; the contributing area to the dam is **571 km²**. Elevations range from about 80 m a.s.l. at the dam to nearly 1,970 m at the highest divide, giving a strongly mountainous, high-relief catchment with a dense, steep drainage network (Figure 1). Physiographic parameters derived from the 5 m NSA DTM are given in Table 1.

Wadi Majlas Catchment - Topography and Drainage

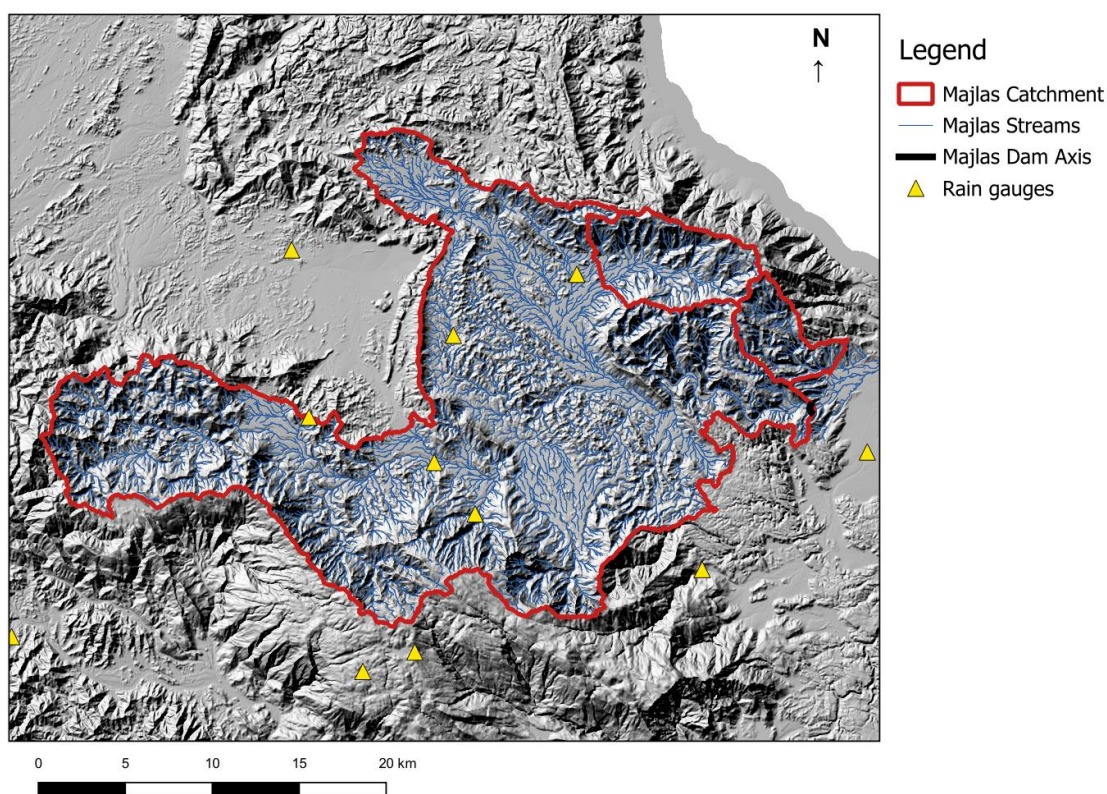


Figure 1 Wadi Majlas catchment - topography, drainage network and rain gauges (QGIS; hillshade from DTM_5m).

Table 1 Physiographic characteristics of the Wadi Majlas catchment (DTM_5m).

Physiographic parameter	Value
Contributing area to dam	571 km ²
Catchment-average curve number (CN)	89
Time of concentration, T _c	400 min
Longest flow path length	68.4 km
Longest flow-path slope	0.0216 m/m

Physiographic parameter	Value
10-85% flow-path length	51.3 km
10-85% flow-path slope	0.0088 m/m
Mean basin slope	0.546 m/m
Basin relief	1,948 m
Elongation ratio	0.394
Drainage density	0.161 km/km ²

2.1. METEOROLOGY

The regional climate is hot arid, with intense but infrequent rainfall from tropical cyclones and convective storms. Monthly air-temperature statistics for the representative Nakhal station are shown in Figure 2 (mean 24-35 C, recorded maxima above 50 C). The wind regime at the dam site, from 20 years of ERA5-Land 10 m wind (2001-2021, Figure 3), is bimodal - drainage winds from the NNE and SSW - with a mean speed of about 1.8 m/s. These support the PMP moisture-maximisation and the reservoir evaporation assessment.

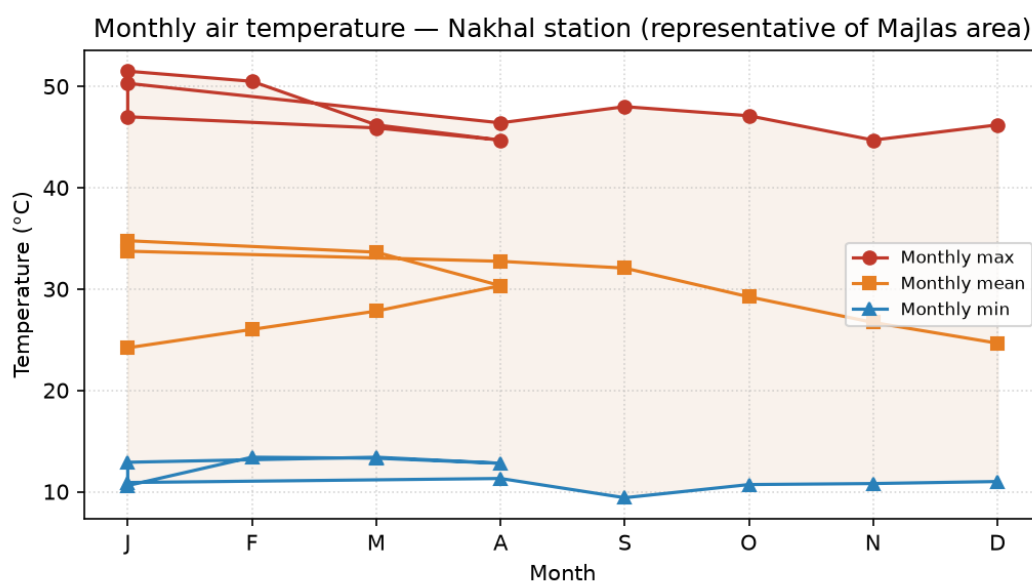


Figure 2 Monthly air temperature at Nakhal station (representative of the Majlas area).

Wind rose at Wadi Majlas dam axis (ERA5-Land 10 m wind, 2001-2021)

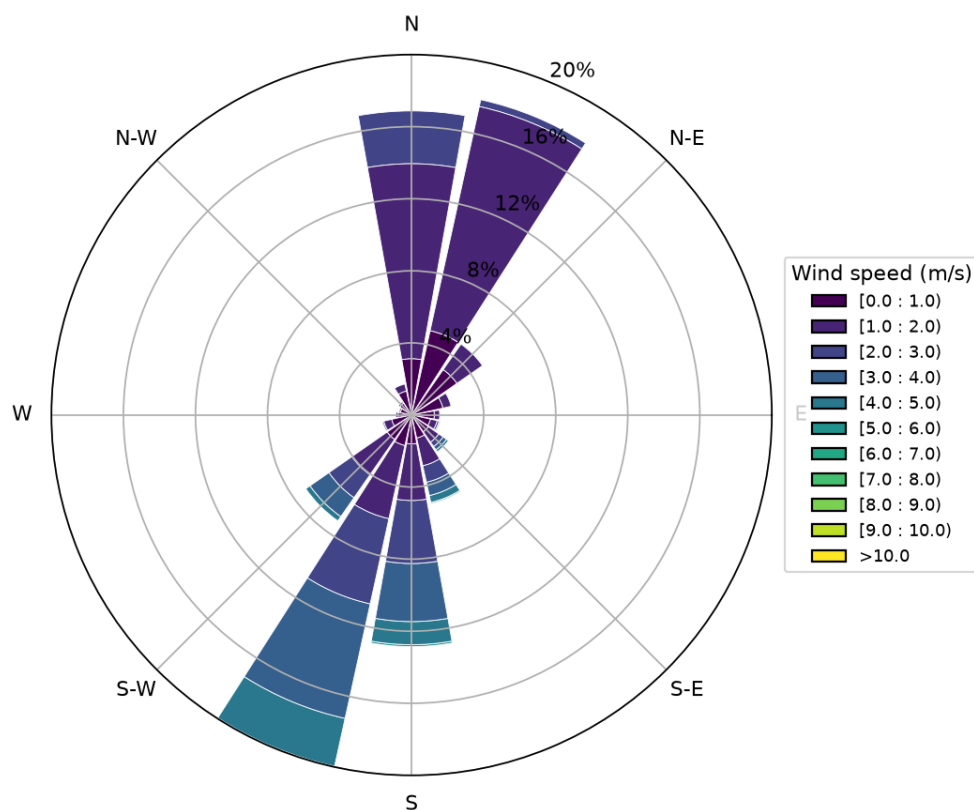


Figure 3 Wind rose at the Wadi Majlas dam axis (ERA5-Land 10 m wind, 2001-2021).

3. AVAILABLE DATA AND RAINFALL RECORDS

The analysis draws on sub-daily rainfall from nine gauges in and around the catchment, the Oman Flood-Mapping regional design rainfall, the 5 m NSA DTM, the GCN250 curve-number grid, reservoir drone-survey hypsometry, and observed cyclone hydrographs (Gonu 2007, Phet 2010). The rain-gauge inventory is given in Table 2; records span 1994-2024, with the three in-catchment (Majlas) gauges - Buei, Taba and Hayfadh - providing the longest series.

Table 2 Rain-gauge inventory and 24-hour annual-maximum statistics.

Gauge	Elev (m)	Catchment	Period	N (yr)	Mean 24h (mm)	Max 24h (mm)
Jabal Abyadh East	1690	Dayqah	1994-2010	13	108	625
Jabal Al Aswad	0	-	2024-2024	1	139	139
Sarin near Habubiyah	445	Mayh	2016-2019	4	58	137
Buei near Buei	450	Majlas	1998-2019	20	53	379
Siya at Siya	371	Mayh	2016-2023	7	31	75
Taba at Taba	310	Majlas	2000-2021	21	58	518
Hayfadh at Hayfadh	230	Majlas	2000-2023	22	52	621
Qurayat at Qurayat	20	Dayqah	2004-2023	19	47	229
Abayah near Misfah	623	Mayh	2016-2022	6	46	101



4. STATISTICAL ANALYSIS OF RAINFALL DATA

4.1. DATA QUALITY ANALYSIS AND OUTLIER TESTING

Annual maximum series (AMS) were extracted for durations from 10 minutes to 24 hours by placing each gauge record on a fixed 10-minute grid and taking the maximum rolling depth in each calendar year of record. Inter-station consistency was checked by double-mass analysis of the 24-hour AMS against the regional mean (Figure 4); the plots are essentially linear, indicating consistent records with no significant breaks. Each series was screened for high outliers with the Grubbs-Beck test (log-domain, 5%). The largest annual maxima - the 2007 Cyclone Gonu depths of 500-625 mm - were **not** flagged as statistical outliers, confirming they are genuine extreme observations to be retained rather than errors to be removed.

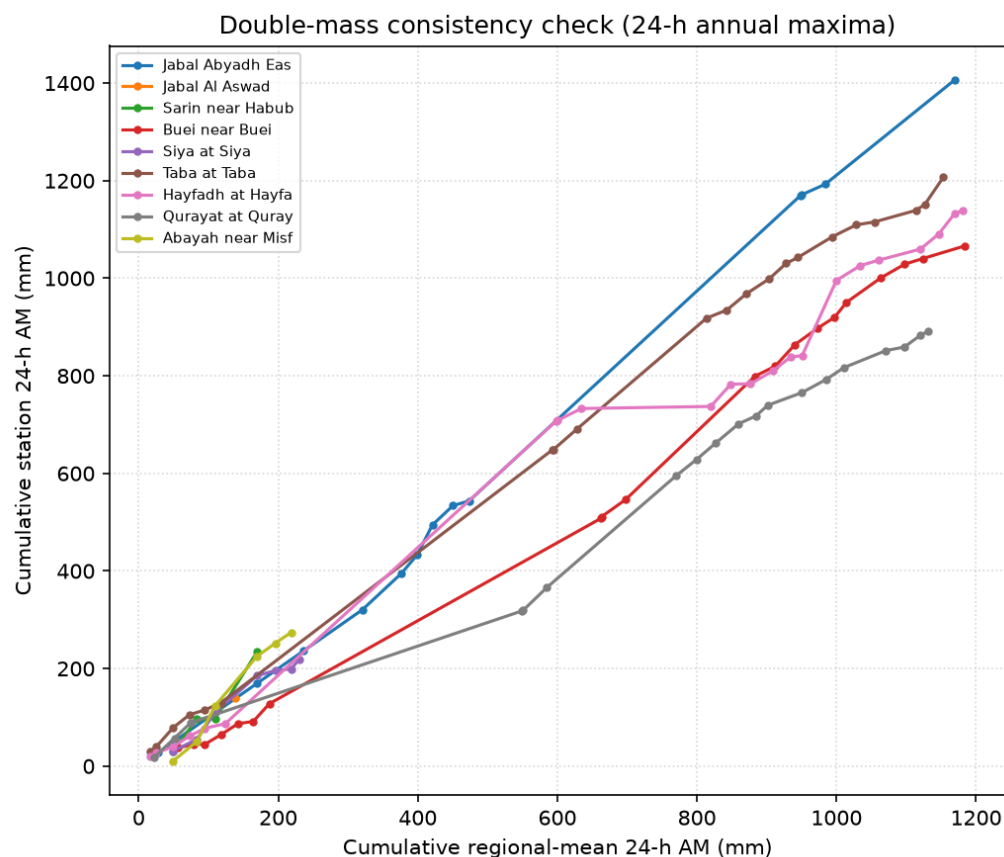


Figure 4 Double-mass consistency check of the 24-hour annual maxima.



4.2. GAP FILLING

Internal gaps in each station's operational record (missing years between the first and last year of record) were infilled by the **normal-ratio method**. For a missing station-year, the estimate is the station's long-term mean multiplied by the average of the concurrent observed ratios (observed maximum / long-term mean) at the neighbouring gauges recording that year:

$$P_{s,y} = N_s \frac{1}{m} \sum_{i=1}^m \frac{P_{i,y}}{N_i}$$

Two rules avoid the errors identified in the earlier analysis: (i) the infilling is performed in a **single pass using observed values only** - filled values are never used to fill other gaps; and (ii) **recorded zeros are kept as real zeros**, not treated as missing. Filling was applied only where at least two neighbouring gauges recorded data, and only to internal gaps (records were not extrapolated before a gauge was installed). This filled 8 of the used station-years (about 7% of the record) and produced balanced series for the regional analysis while leaving the observed extremes unchanged.

4.3. REGIONAL FREQUENCY ANALYSIS (HOSKING & WALLIS)

Design rainfall was estimated by the **index-flood regional frequency analysis of Hosking & Wallis (1997)**. In this approach each site AMS is standardised by its at-site mean (the index rainfall) and pooled into a common regional dimensionless growth curve, so that the effective record length - and hence the reliability of rare quantiles - is increased. The pooled data are fitted with the **Generalised Extreme Value (GEV) distribution using L-moments**. Only the GEV/L-moment estimator is used, because L-moments are far less sensitive than product moments to the few dominant cyclone maxima, the GEV is the asymptotic distribution for annual maxima and the standard for regional rainfall analysis, and a single estimator keeps the procedure transparent and reproducible.

The three formal H&W measures were computed for the region (the five gauges with at least eight years of record). **Discordancy** (Table 3): no site exceeds the critical value, so the region is internally consistent. **Heterogeneity** $H = -1.139 (< 1)$: the region is statistically homogeneous - the gauges share the same, cyclone-dominated frequency behaviour. **Goodness-of-fit**: the standard kappa-based Z test is not applicable because the extreme regional L-skewness (0.69, driven by Cyclone Gonu) lies at the edge of the kappa domain; the GEV L-kurtosis for the fitted shape (0.56) is close to



the observed regional value (0.62), so the GEV is the most appropriate standard distribution and is adopted, while noting that no three-parameter distribution fully captures the Gonu-dominated upper tail.

Table 3 Hosking & Wallis discordancy (Di) and L-moment ratios by gauge (critical Di = 1.333).

Gauge	N	L-CV	L-skew	L-kurt	Di	Discordant?
Jabal Abyadh East	17	0.63	0.647	0.559	1.17	no
Buei near Buei	22	0.647	0.668	0.598	0.216	no
Taba at Taba	22	0.668	0.73	0.667	1.3	no
Hayfadh at Hayfadh	23	0.747	0.774	0.726	1.125	no
Qurayat at Qurayat	19	0.554	0.591	0.518	1.19	no

4.4. REGIONAL GROWTH CURVE AND DESIGN DEPTHS

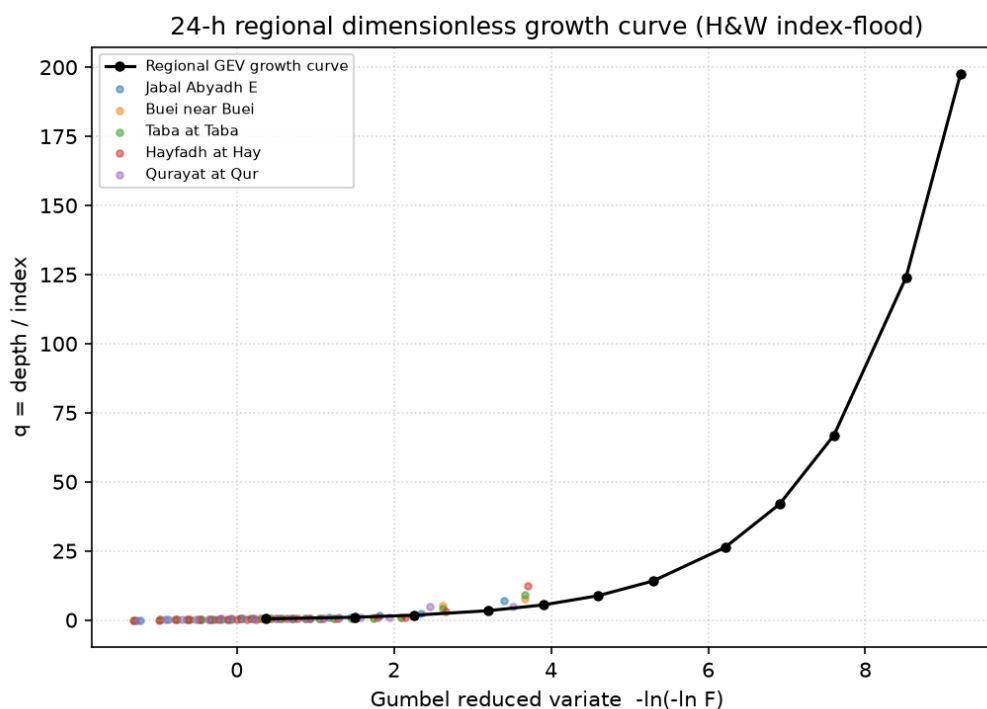


Figure 5 24-hour regional dimensionless growth curve (Hosking & Wallis index-flood).

The regional growth curve (Figure 5) applied to the study-area index rainfall (record-length-weighted mean of the at-site means, 57.6 mm) gives the representative 24-hour design depths in Table 4.



Table 4 Representative 24-hour design rainfall depths (local H&W RFA, GEV/L-moments).

Return period (yr)	24-h depth (mm)
2	24
5	58
10	100
25	194
50	315
100	507
200	813
500	1,512
1000	2,414
2000	3,852
5000	7,139
10000	11,382

4.5. INTENSITY-DURATION-FREQUENCY (IDF) CURVES

Station-based IDF curves were derived for each gauge, and a regional IDF from the pooled data, by applying the regional growth curve to the index rainfall of each duration (10 min-24 h). Because a single growth curve is used, the return-period curves stay parallel and do not cross or diverge beyond one hour - the instability seen when each duration is fitted independently. The station-based IDF curves are shown in Figure 6 and the regional IDF in Figure 7.



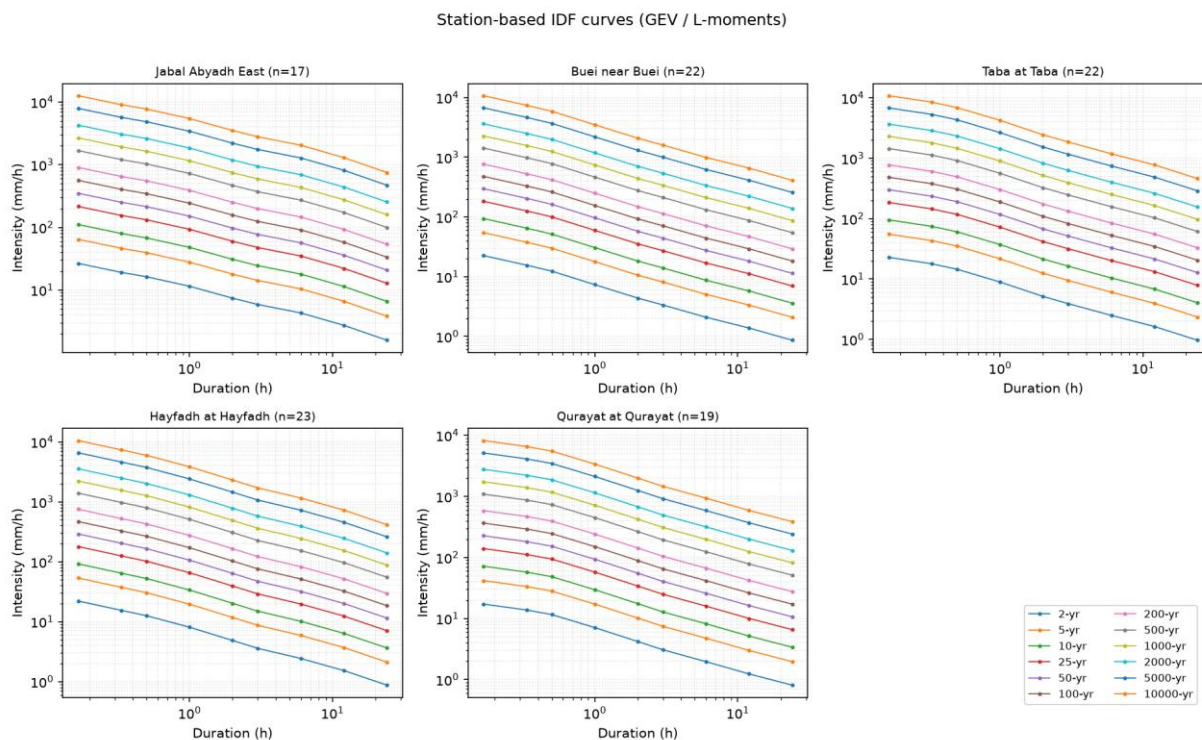


Figure 6 Station-based IDF curves (GEV/L-moments, index-flood).

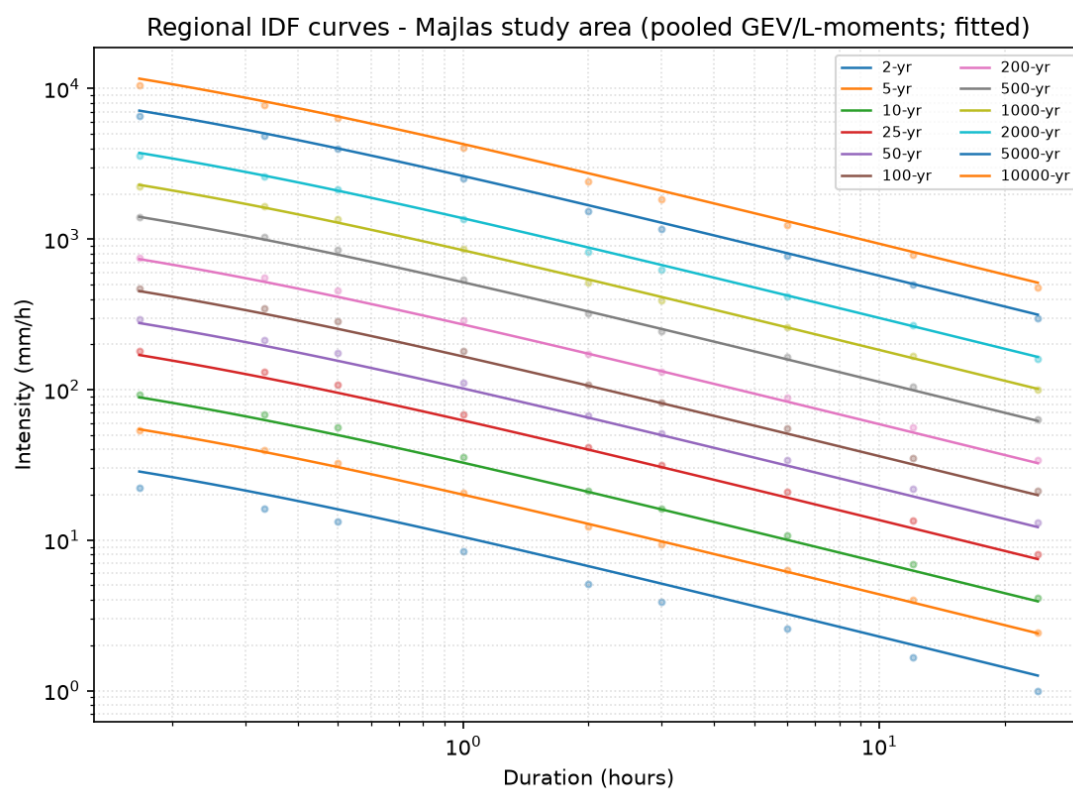


Figure 7 Regional IDF curves for the Wadi Majlas study area (pooled data; fitted).

The regional IDF is well described ($R^2 = 0.9989$ in log space) by the Sherman-type equation $i = 111.776 T^{0.7059} / (D + 5.0428)^{0.684}$, with intensity i in mm/h, return period T in years and duration D in minutes.



5. COMPARISON WITH THE OMAN FLOOD-MAPPING RAINFALL

The Oman Flood-Mapping study provides regional 24-hour design rainfall covering the Majlas catchment; these depths drove all HEC-HMS design runs. Table 5 and Figure 8 compare them with the local H&W depths.

Table 5 24-hour design rainfall: local H&W RFA versus Oman Flood-Mapping.

Return period (yr)	Local H&W (mm)	Flood Mapping (mm)	Ratio
2	24	28	0.87
5	58	55	1.05
10	100	106	0.95
25	194	138	1.41
50	315	168	1.87
100	507	205	2.47
200	813	251	3.24
500	1,512	327	4.62
1000	2,414	400	6.04
2000	3,852	489	7.88
5000	7,139	639	11.17
10000	11,382	685	16.61

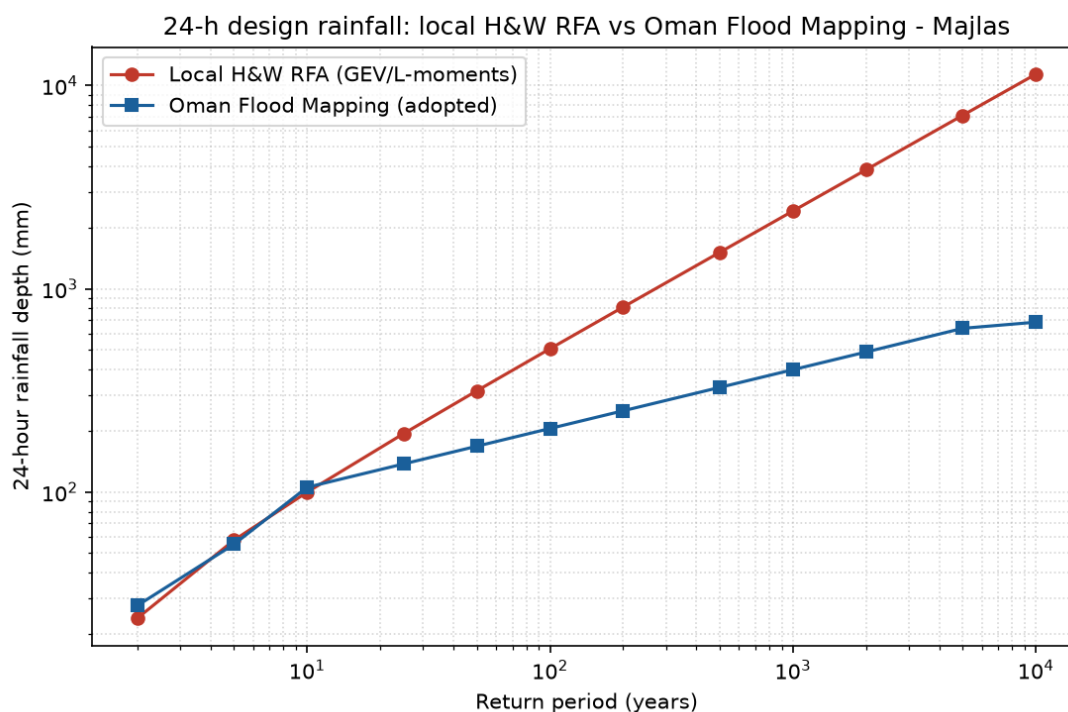


Figure 8 24-hour design rainfall - local H&W RFA versus Oman Flood-Mapping.

The two data sets are almost identical at the 2-year event and within about 25% to the 10-year event. Beyond this the local analysis diverges upward, reaching about three times the Flood-Mapping depth at 100 years. This is the signature of the short local records being dominated by a single very rare event (Cyclone Gonu, 2007), which produces a heavy-tailed growth curve on extrapolation far beyond the observation period. The Flood-Mapping depths, drawing on a longer, spatially extensive database, are more robust at rare return periods and were therefore **adopted as the design rainfall**; the local analysis serves as an independent check that confirms agreement in the well-observed range and quantifies the sensitivity of the tail to the Gonu event.



6. PROBABLE MAXIMUM PRECIPITATION (PMP)

The PMP is the greatest depth of precipitation meteorologically possible for a given duration over the basin (WMO, 2009). The deterministic moisture-maximisation and storm-transposition approach of the regional Oman PMP study was adopted. Cyclone Gonu (2007), the highest recorded regional 24-hour rainfall, was maximised and transposed to the project area, the storm rainfall being scaled by the ratio of precipitable water under maximum and observed dew-point conditions:

$$PMP = R_0 \frac{W_m}{W_s}$$

ERA-5 dew-point temperatures (1940-present) provided the maximising factors. The resulting 24-hour PMP grid over the catchment (Figure 9) ranges from about 780 mm at the dam to over 1,600 mm on the highest ground; the design value adopted from the regional PMP grid is consistent with these magnitudes.

24-h Probable Maximum Precipitation - Wadi Majlas

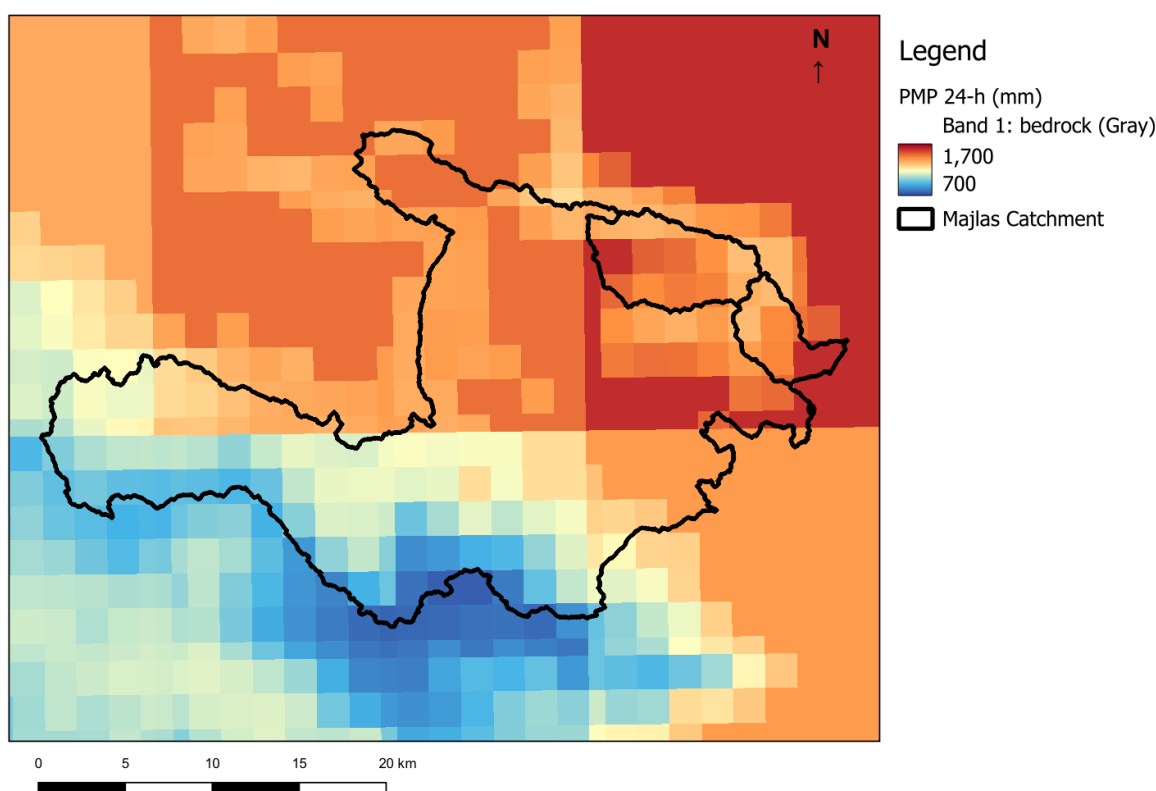


Figure 9 24-hour Probable Maximum Precipitation over the Wadi Majlas catchment (QGIS).

7. RAINFALL-RUNOFF MODEL

The catchment response was simulated in HEC-HMS with the SCS curve-number loss method, the SCS unit-hydrograph transform with Kirpich lag times and reach routing. Design storms were 24-hour hyetographs with the Flood-Mapping temporal distribution and a depth-area reduction factor.

7.1. SCS CURVE-NUMBER LOSS METHOD

Direct runoff is computed from the SCS relationship:

$$Q = \frac{(P - I_a)^2}{(P - I_a + S)}$$

where Q is runoff, P is rainfall, I_a is initial abstraction and S the potential maximum retention, with

$$S = \frac{25400}{CN} - 254 \text{ (mm)} \quad I_a = 0.2 S$$

Arid- and semi-arid rangeland curve numbers follow the Oman Hydrological Design Standards (OHDS-2017); for the mountainous catchment the applicable values are CN 86-90 (slope < 18%) and 90-94 (slope > 18%).

7.2. CURVE NUMBER GRID

Curve numbers were taken from the GCN250 global product (250 m; ESA CCI land cover with HYSOGs250m soil groups) and clipped to the catchment (Figure 10). The catchment-average CN is about **89** (AMC II); sub-basin values range 86.5-91.0 (Table 8).



SCS Curve Number Grid (GCN250) - Wadi Majlas

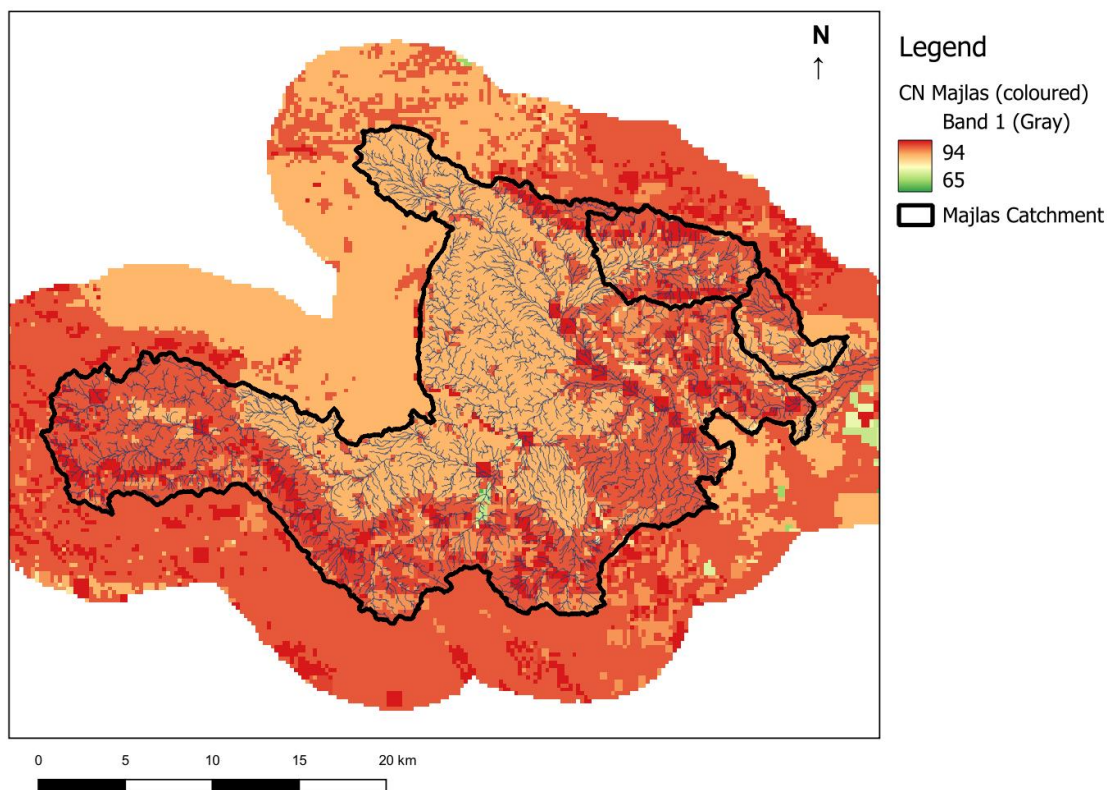


Figure 10 SCS curve-number grid (GCN250) over the Wadi Majlas catchment (QGIS).

7.3. AREA REDUCTION FACTOR

Point rainfall over-estimates the average depth over a large catchment; an Area Reduction Factor (ARF) from the extreme Oman storms was applied through the depth-area analysis in the final model:

$$ARF = 1 - 0.001 A^{(1.25 - 0.062 \ln A)}$$

Table 6 Area reduction factor applied in the depth-area analysis.

Area (km2)	Area reduction factor
1	1.000
5	0.994
10	0.987
20	0.976
30	0.965
40	0.956
50	0.948

Area (km2)	Area reduction factor
100	0.913
200	0.864
500	0.776
1000	0.694
2000	0.606

7.4. CATCHMENT, STREAMS AND SUB-BASINS

Catchment and stream delineation used the 5 m DTM after sink-filling, flow-direction and flow-accumulation processing with a 1 km² minimum drainage-area threshold. The HEC-HMS sub-basin delineation is shown in Figure 11. Sub-basin lag times were computed from the Kirpich formula and initial abstractions from the curve number:

HEC-HMS Sub-basin Delineation - Wadi Majlas

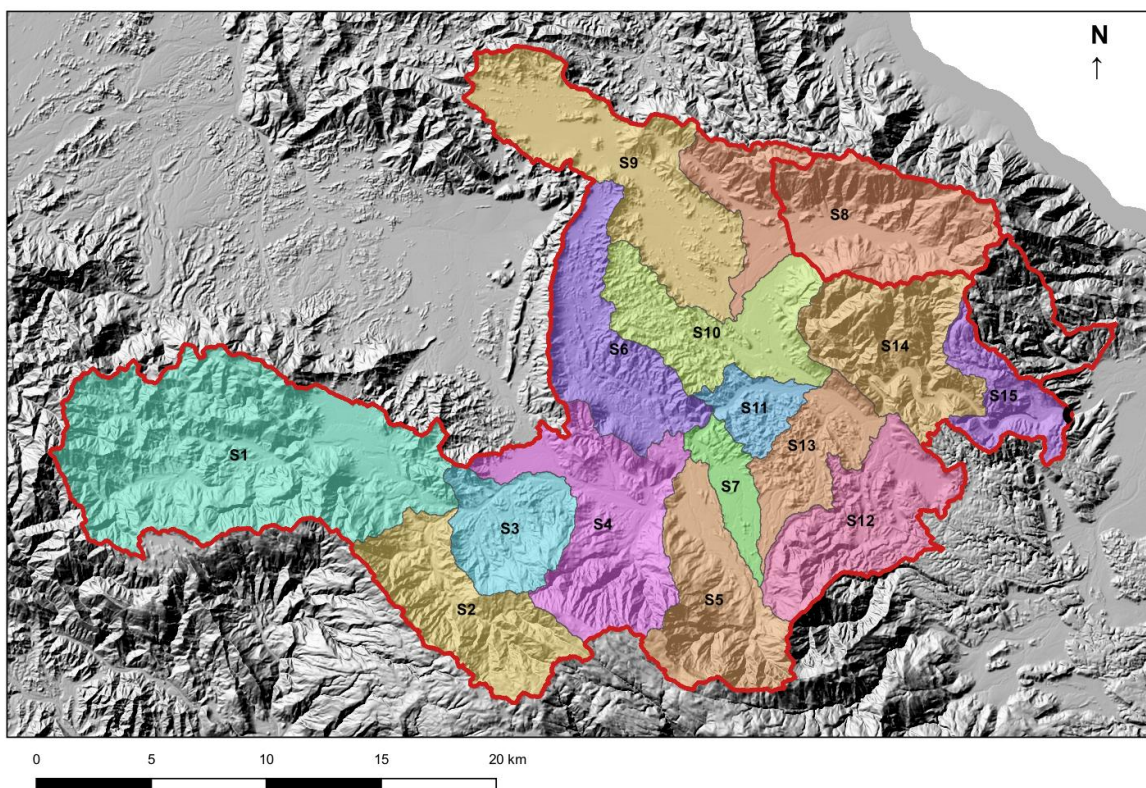


Figure 11 HEC-HMS sub-basin delineation of the Wadi Majlas model (QGIS).

$$t_{lag} = 0.6 \frac{1}{52} \frac{(L_{10-85} \times 1000)^{0.77}}{S^{0.385}}$$

where L_{10-85} is the 10-85% flow-path length (km) and S its slope. The 20 model sub-basins - 15 upstream of the dam plus the downstream sub-basins for the dyked channel - are summarised in Table 8.

Table 7 Characteristics of the HEC-HMS sub-basins (Wadi Majlas model).

Sub-basin	Area (km ²)	CN	Ia (mm)	Lag (min)	Downstream
S1	106.8	90.3	5.45	110	J1
S2	36.6	91	5.04	34.8	J1
S3	24.2	87.7	7.12	44	J2
S4	47.9	88	6.94	80.4	J3
S5	36.8	90	5.63	46.9	J3
S6	36.7	86.5	7.94	113	J4
S7	10.7	86.8	7.76	60.8	J4
S9	56.5	86.5	7.92	147	J5
S10	34.2	87.7	7.13	92.3	J6
S11	10.6	87.6	7.21	46.2	J6
S12	37.0	90.5	5.36	65.5	J7
S13	21.6	89.8	5.78	64.6	J7
S14	37.3	90	5.62	70.6	J8
S15	15.9	89.4	6.05	30.3	Majlas Dam
Subbasin-1	40.4	90.1	5.55	43.2	J9
Subbasin-2	18.0	89.2	6.17	43.5	J5
Sdn1	3.0	88	6.92	28.1	J10
Sdn2	20.3	88.5	6.58	24.9	J11
Sdn3	3.5	88.6	6.52	25.7	O1
Sdn4	6.2	88.4	6.65	33.6	O1
TOTAL	604.3				



8. MODEL CALIBRATION AND REVIEW OF THE PREVIOUS STUDY

The model was calibrated against observed hydrographs from Cyclone Gonu (June 2007) - the largest instrumented event in the region - and checked against Cyclone Phet (2010). The previous study calibrated the model to observed discharge hydrographs at the two gauges near the dam, Hayfadh and Majlass. The recorded 24-hour Gonu depths at the contributing gauges are listed in Table 9.

On re-calibration, the Hayfadh gauge hydrograph was reproduced well with consistent, physically reasonable parameters (as also reported previously), but the observed hydrograph at the Majlass gauge could not be matched with the gauged Gonu rainfall. A sensitivity analysis progressively reduced the recorded storm rainfall to find the reduction needed to reproduce that hydrograph. Matching it required reducing the gauged 24-hour rainfall to physically implausible levels at several stations - for example 384 to 38 mm at Buei and 231 to 23 mm at Qurayat - while others required almost no change (Table 9).

Table 8 Rainfall reduction required to reproduce the questioned Majlass observed hydrograph (Gonu 2007).

Gauge	Gonu 24-h (mm)	Reduced to match (mm)	Reduction
Jabal Abyadh East	629	252	-60%
Buei near Buei	384	38	-90%
Taba at Taba	518	414	-20%
Hayfadh at Hayfadh	625	625	0%
Qurayat at Qurayat	231	23	-90%

Such non-uniform, extreme reductions have no physical basis. The conclusion is that the **observed hydrograph at the Majlass gauge is erroneous** - most likely affected by a rating or gauging error at high stage - and should not have been used as a calibration target; this was not identified in the previous report. The present model was calibrated to the reliable Hayfadh hydrograph and to observed peak flows, validated against Cyclone Phet, and the questioned Majlass record was excluded. The adopted parameters are the GCN250 curve numbers (catchment average about 89) and Kirpich lag times of Table 8.



9. DESIGN FLOOD RESULTS

Design floods were computed for return periods from 2 to 10,000 years and the PMP, using the Flood-Mapping design rainfall and the calibrated model, in two configurations: **without** reservoir storage (dam sites as flow nodes: Majlas and Haifaz), and **with** the Majlas reservoir storage and spillway routed.

The Majlas reservoir has a full-supply level of 83 m and crest of 94 m, giving about 94 Mm³ of storage (dam height about 74.6 m); its elevation-area-storage relationship is shown in Figure 12.

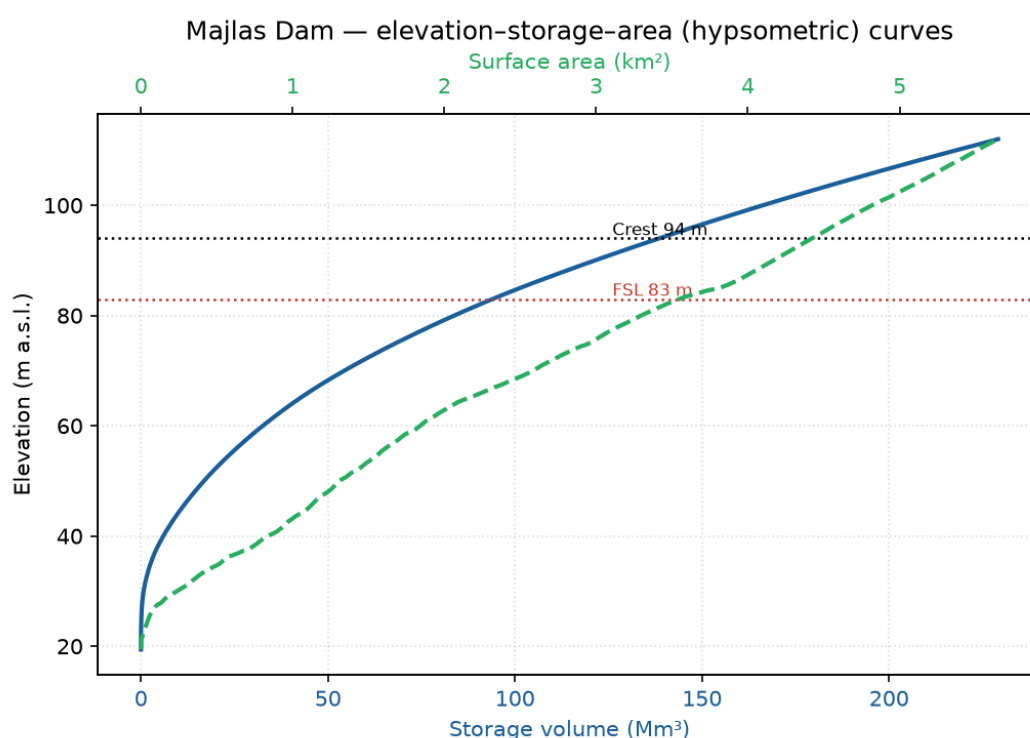


Figure 12 Majlas Dam elevation-storage-area (hypso-metric) curves.

9.1. PEAK DISCHARGES

Table 10 gives peak discharges at key locations; Figure 13 shows the outlet peak-frequency curve for both configurations.

Table 9 Design peak discharges (m³/s) at key locations, with and without the Majlas reservoir.

Location / configuration	2-yr	10-yr	100-yr	1,000-yr	10,000-yr	PMP
Majlas Dam (inflow) - no res.	92	1,161	3,452	8,289	14,166	18,726



Location / configuration	2-yr	10-yr	100-yr	1,000-yr	10,000-yr	PMP
Catchment outlet O1 - no res.	86	1,126	3,351	8,026	13,798	18,692
Downstream J10 - no res.	89	1,144	3,398	8,131	13,923	18,532
Downstream J12 - no res.	88	1,135	3,394	8,139	13,968	18,751
Majlas Dam (routed out) - with res.	0	0	0	2,228	8,749	17,224
Catchment outlet O1 - with res.	17	216	449	2,180	8,585	17,286
Downstream J10 - with res.	2	29	56	2,211	8,665	17,123
Downstream J12 - with res.	17	221	430	2,198	8,664	17,303

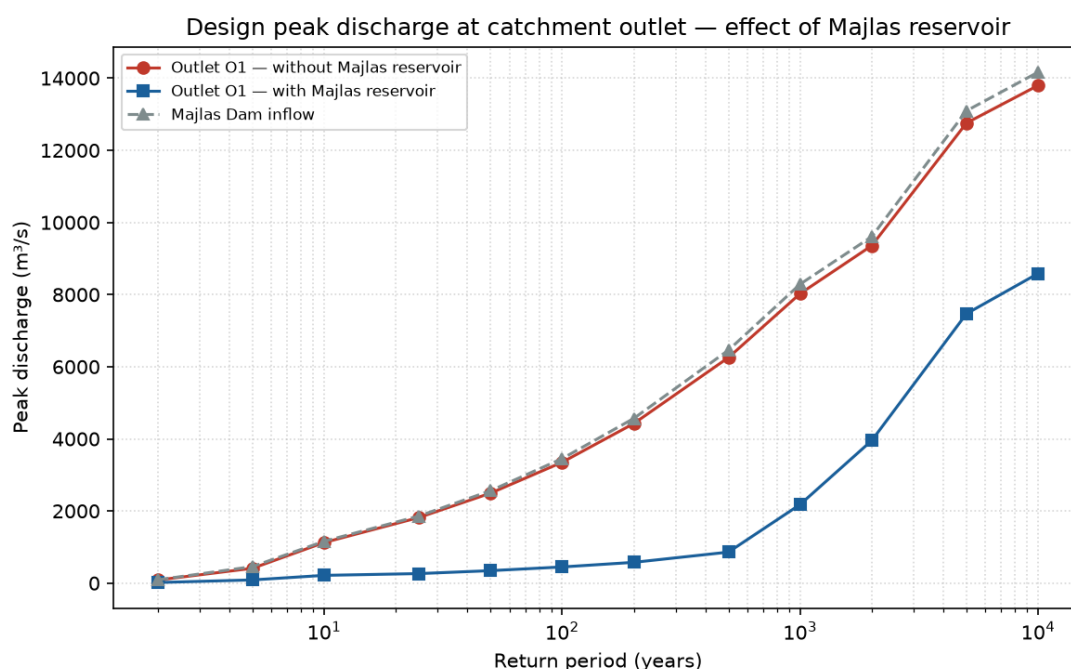


Figure 13 Peak-discharge frequency at the catchment outlet - effect of the Majlas reservoir.

9.2. RESERVOIR ROUTING AND ATTENUATION

The reservoir stores the full inflow up to about the 200-year event (zero spillway outflow) and strongly attenuates larger floods (Table 11, Figures 14-15). The 10,000-year peak is reduced from 14,166 to 8,749 m³/s (38%); even for the PMP the routed outflow (17,224 m³/s) stays below the inflow (18,726 m³/s).

Table 10 Majlas reservoir routing - peak inflow, routed outflow and attenuation.

Return period	Peak inflow (m ³ /s)	Peak outflow (m ³ /s)	Attenuation
2-yr	92	0	100%



Return period	Peak inflow (m3/s)	Peak outflow (m3/s)	Attenuation
5-yr	460	0	100%
10-yr	1,161	0	100%
25-yr	1,859	0	100%
50-yr	2,561	0	100%
100-yr	3,452	0	100%
200-yr	4,566	0	100%
500-yr	6,459	877	86%
1000-yr	8,289	2,228	73%
2000-yr	9,607	4,029	58%
5000-yr	13,095	7,608	42%
10000-yr	14,166	8,749	38%
PMP	18,726	17,224	8%

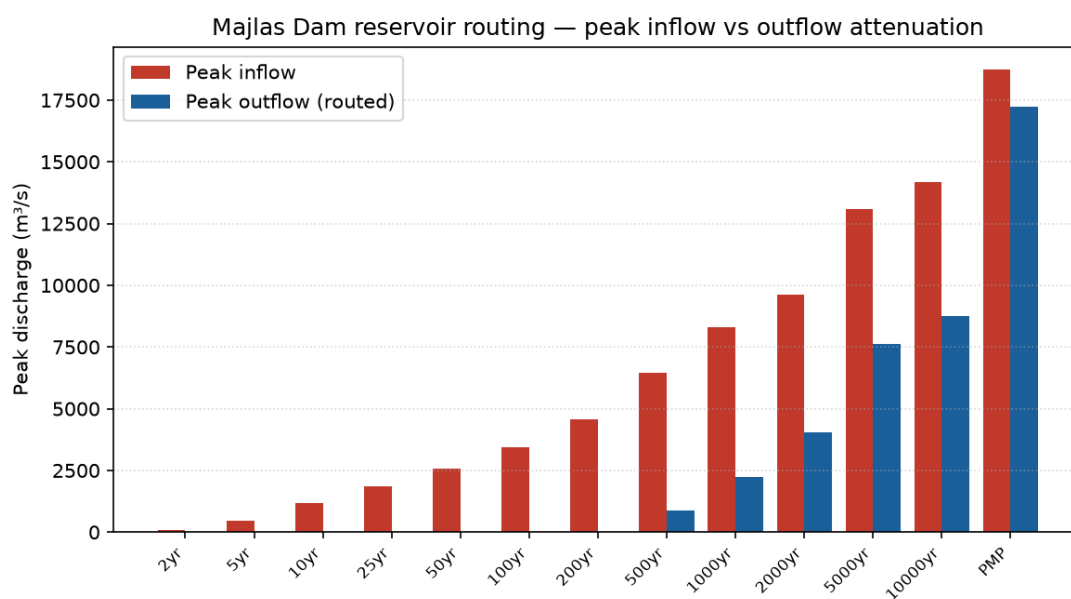


Figure 14 Majlas reservoir routing - peak inflow versus routed outflow.



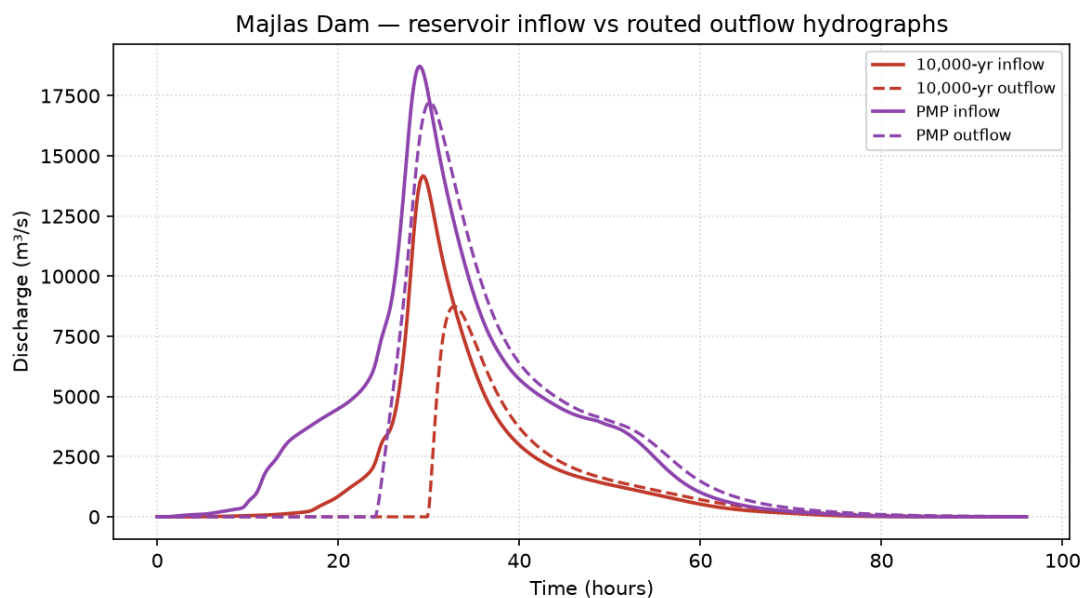


Figure 15 Majlas reservoir inflow and routed-outflow hydrographs (10,000-year and PMP).

The downstream peaks (J10, J12 and outlet O1 in Table 10) are the design flows for the dyked channel. Because the reservoir removes the upstream contribution up to moderate return periods, the downstream design flows under the with-reservoir configuration are governed by the local downstream sub-catchments to about the 200-year event and by reservoir spill for rarer events.

10. CONCLUSIONS AND RECOMMENDATIONS

- The catchment to the dam is 571 km² (604 km² with the downstream sub-catchments), mountainous and steep, with a catchment-average curve number of about 89.
- The rainfall records are consistent (double-mass) and the cyclone maxima are genuine extremes (not outliers); internal gaps were filled by the normal-ratio method without reusing filled values or mis-treating real zeros.
- The region is statistically homogeneous ($H = -1.1$; no discordant sites). Local H&W GEV/L-moment design rainfall matches the Oman Flood-Mapping rainfall to the 10-year event but diverges at rare return periods because of Cyclone Gonu dominance; the Flood-Mapping depths were adopted for design.
- The observed Majlass Gonu hydrograph used for calibration in the previous study is erroneous (reproducible only with implausible rainfall reductions) and was excluded; this was not flagged previously.
- The Majlas reservoir stores the full inflow to about the 200-year event and attenuates the 10,000-year peak by about 38% (14,166 to 8,749 m³/s). Design peaks and hydrographs (2-10,000 years and PMP) are provided for the dam spillway and the downstream dyked channel.

It is recommended that design proceed on the Flood-Mapping-based design floods with reservoir routing, that the questioned historical gauge record be formally set aside, and that the local rainfall analysis be revisited as additional years of record accumulate.



11. REFERENCES

Hosking, J.R.M. & Wallis, J.R. (1997). Regional Frequency Analysis: An Approach Based on L-Moments. Cambridge University Press.

Dalrymple, T. (1960). Flood-Frequency Analyses. USGS Water-Supply Paper 1543-A.

Hosking, J.R.M. (1990). L-moments: analysis and estimation of distributions using linear combinations of order statistics. J. Royal Stat. Soc. B, 52(1), 105-124.

World Meteorological Organization (2009). Manual on Estimation of Probable Maximum Precipitation (PMP), WMO-No. 1045.

Jaafar, H.H. et al. (2019). GCN250, new global gridded curve numbers for hydrologic modeling and design. Scientific Data 6, 145.

Oman Hydrological Design Standards (OHDS-2017), Sultanate of Oman.

US Army Corps of Engineers (2000). Hydrologic Modeling System HEC-HMS Technical Reference Manual.

